

BRAC UNIVERSITY
Department of Computer Science and Engineering

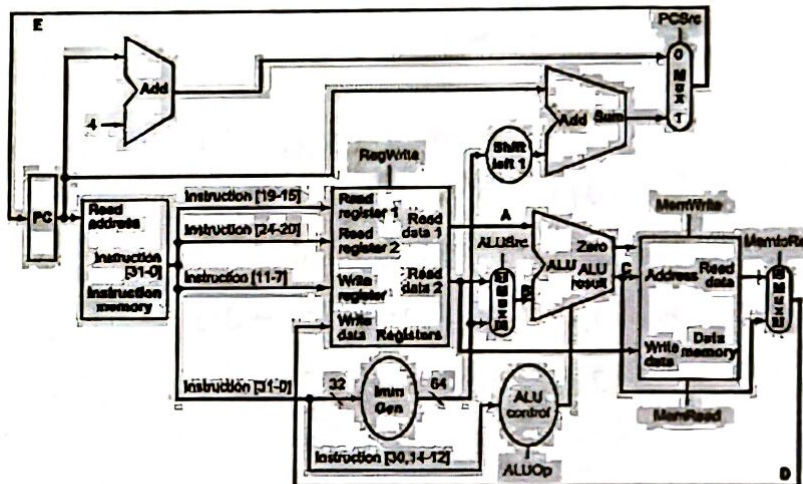
Examination: Surprise Test - 4
 Duration: 10 minutes

Semester: Spring 2025
 Full Marks: 05

CSE 340: Computer Architecture

Name: <u>Solution</u>	ID:	Section:
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1.



Determine the values of the following control bits when executing the instruction "Add X21, X22, X23" in the provided single-cycle datapath.

Branch	ALUSrc	RegWrite	MemToReg
0	1	1	1

2. **Identify** the necessary resources or components from the list provided to construct the datapath for each of the following instructions. Just put a tick(✓) sign in the respective cells to show your selection.[3]

		PC	Instruction Memory	Data Memory	Register File	ALU	Immediate Generation Unit	Control Unit	ALU Control
i	ADD X21, X22, X23	✓	✓		✓	✓		✓	✓
ii	Slli X21, X22, 23	✓	✓		✓	✓	✓	✓	✓
iii	SB X21, 10(X22)	✓	✓	✓	✓	✓	✓	✓	✓